

# A COUNTEREXAMPLE TO A CONJECTURE OF FRANKLÍN ON (321, 1342)-AVOIDERS

ABSTRACT. Franklín conjectured that the number of indecomposable permutations avoiding 321 and 1342 with exactly  $k$  inversions is  $1 + \binom{k+1}{2}$ . The case  $k = 1$  already gives a counterexample. Using a disjoint structural cover of  $\text{Av}(321, 1342)$ , we determine the bivariate generating function of the indecomposable members by length and inversions and prove that the exact count is  $1 + \binom{k}{2}$ , so the conjectured sequence is the correct one shifted by one in the index. The number of indecomposable avoiders of length  $n$  is  $1 + \binom{n}{3}$ .

## 1. EXACT ENUMERATION

For a set  $B$  of patterns, let  $\text{Av}(B)$  denote the permutations avoiding every pattern in  $B$ , and let  $\text{inv}(\pi)$  be the inversion number of  $\pi$ . A nonempty permutation  $\pi = \pi_1 \cdots \pi_n$  is *indecomposable* if no proper prefix  $\pi_1 \cdots \pi_j$  has value set  $\{1, \dots, j\}$ . Define

$$I_k = \{\pi \in \text{Av}(321, 1342) : \pi \text{ is indecomposable and } \text{inv}(\pi) = k\},$$

and let  $S_n^*$  be the indecomposable members of  $\text{Av}(321, 1342)$  of length  $n$ . These are the sets written  $I_k(321, 1342)$  and  $S_n^*(321, 1342)$  in [2], where the bare symbols  $I_k$  and  $S_n^*$  denote the corresponding sets of all indecomposable permutations; we use the abbreviated symbols throughout. As recorded in [2], an indecomposable permutation of length  $n$  has at least  $n - 1$  inversions, so  $I_k$  consists of permutations of length at most  $k + 1$  and is in particular finite; this is what makes enumeration by inversions possible at all. For the members of  $\text{Av}(321, 1342)$  the same bound is recovered in the proof of Theorem 1.

In an unnumbered remark in the introduction of [2, p. 3], offered on the basis of numerical evidence, Franklín writes: “Another case is  $I_k(321, 1342)$ , which we conjecture to have  $k(k + 1)/2 + 1$  elements.” In the abbreviated notation this is the assertion

$$|I_k| = 1 + \binom{k+1}{2}.$$

By Theorem 1, the conjectured value  $1 + \binom{k+1}{2}$  is precisely  $|I_{k+1}|$ , so the conjectured sequence is the correct one shifted by one in the index. The

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shift is not a difference of convention: in [2] as here,  $k$  is the number of inversions of the permutation and  $I_0 = \{1\}$ .

A permutation with one inversion is obtained from the identity by interchanging two adjacent consecutive values, and it is indecomposable only in the case 21. Hence  $I_1 = \{21\}$ , whereas the conjectured formula gives  $|I_1| = 2$ .

The same passage of [2] reports experimental evidence that  $S_n^*(321, 1342)$  grows like  $n^3/6$ ; the second formula of Theorem 1 confirms that prediction and sharpens it to the closed form  $1 + \binom{n}{3}$ . The asymptotic bound on  $S_n^*(321, 1342)$  drawn there from the conjectured count is unaffected by the correction, since  $1 + \binom{k}{2}$  and  $1 + \binom{k+1}{2}$  are both asymptotic to  $k^2/2$ . The following theorem gives the exact correction.

**Theorem 1.** *Let*

$$J(x, q) = \sum_{\substack{\pi \in \text{Av}(321, 1342) \\ \pi \text{ indecomposable}}} x^{|\pi|} q^{\text{inv}(\pi)}.$$

*Then*

$$(1) \quad J(x, q) = \frac{x}{1 - xq} + \frac{x^3 q^2}{1 - xq} \sum_{a \geq 0} \frac{(xq)^a}{(1 - xq^{a+1})(1 - xq^{a+2})}.$$

*Consequently, for every  $k \geq 0$  and  $n \geq 1$ ,*

$$|I_k| = 1 + \binom{k}{2}, \quad |S_n^*| = 1 + \binom{n}{3}.$$

*In particular, the conjectured formula is false for every  $k \geq 1$ .*

*Proof.* The disjoint structural cover of Bean, Gudmundsson, and Ulfarsson [1, Proposition 7 and Equation (6)] shows that every nonempty member of  $\text{Av}(321, 1342)$  has a unique representation of one of the following two types. The cover is stated there as a gridded tiling, in which the maximum shares a column with an increasing cell and hence interleaves it; we have unpacked it into the explicit block form below. All displayed blocks may be empty, and inequalities involving blocks are understood elementwise.

In the first type,

$$\pi = \alpha n \beta,$$

where  $\alpha \in \text{Av}(321, 1342)$ , the block  $\beta$  is increasing, and all entries of  $\alpha$  are smaller than all entries of  $\beta$ . If  $\alpha$  is nonempty, its entries are  $1, \dots, |\alpha|$ , so  $\pi$  is decomposable. Thus the indecomposable permutations of this type are

$$n \ 1 \ 2 \ \cdots \ (n-1), \quad n \geq 1,$$

and their contribution to  $J(x, q)$  is

$$\sum_{n \geq 1} x^n q^{n-1} = \frac{x}{1 - xq}.$$

In the second type,

$$\pi = c \gamma_1 \gamma_2 n b \delta_1 \delta_2,$$

where the four blocks are increasing and

$$\gamma_2 < b < \delta_1 < c < \gamma_1 < \delta_2 < n.$$

Every such permutation is indecomposable. Indeed, a proper prefix ending before  $n$  contains  $c$  but omits the smaller entry  $b$ , while a proper prefix of length  $j < n$  containing  $n$  contains a value larger than  $j$ .

Set

$$a = |\gamma_1|, \quad u = |\gamma_2|, \quad v = |\delta_1|, \quad w = |\delta_2|.$$

The length is  $3 + a + u + v + w$ . The inversions are those from  $c$  to  $\gamma_2, b, \delta_1$ , from each entry of  $\gamma_1$  to  $\gamma_2, b, \delta_1$ , and from  $n$  to  $b, \delta_1, \delta_2$ . Hence

$$\text{inv}(\pi) = (u + 1 + v) + a(u + 1 + v) + (1 + v + w) = 2 + a(u + v) + a + u + 2v + w.$$

Therefore the second type contributes

$$\begin{aligned} & \sum_{a,u,v,w \geq 0} x^{3+a+u+v+w} q^{2+a(u+v)+a+u+2v+w} \\ &= \frac{x^3 q^2}{1 - xq} \sum_{a \geq 0} \frac{(xq)^a}{(1 - xq^{a+1})(1 - xq^{a+2})}, \end{aligned}$$

which proves (1).

As a check on the transcription, the two types give, ignoring inversions, the functional equation

$$F(x) = 1 + \frac{x F(x)}{1 - x} + \frac{x^3}{(1 - x)^4}$$

for the generating function  $F$  of  $\text{Av}(321, 1342)$  by length, the summand 1 accounting for the empty permutation. Hence

$$\begin{aligned} F(x) &= \frac{(1 - x)^4 + x^3}{(1 - x)^3(1 - 2x)} = \frac{1 - 4x + 6x^2 - 3x^3 + x^4}{1 - 5x + 9x^2 - 7x^3 + 2x^4} \\ &= 1 + x + 2x^2 + 5x^3 + 13x^4 + 32x^5 + \cdots, \end{aligned}$$

which is the generating function of the class recorded in [1].

For the first type,  $\text{inv}(\pi) = |\pi| - 1$ ; for the second,

$$\text{inv}(\pi) - (|\pi| - 1) = a(u + v) + v \geq 0.$$

Thus setting  $x = 1$  is well defined coefficientwise. For every  $A \geq 0$ ,

$$\sum_{a=0}^A \frac{q^a}{(1 - q^{a+1})(1 - q^{a+2})} = \frac{1}{1 - q} \left( \frac{1}{1 - q} - \frac{q^{A+1}}{1 - q^{A+2}} \right).$$

Letting  $A \rightarrow \infty$  coefficientwise gives

$$\sum_{a \geq 0} \frac{q^a}{(1 - q^{a+1})(1 - q^{a+2})} = \frac{1}{(1 - q)^2}.$$

Consequently,

$$J(1, q) = \frac{1}{1 - q} + \frac{q^2}{(1 - q)^3},$$

and coefficient extraction yields

$$|I_k| = 1 + \binom{k}{2}.$$

Finally, setting  $q = 1$  in (1) gives

$$J(x, 1) = \frac{x}{1 - x} + \frac{x^3}{(1 - x)^4},$$

whose coefficient of  $x^n$  is  $1 + \binom{n}{3}$ . □

#### REFERENCES

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